Project Specifications

Deploying and Securing a Virtual Infrastructure

Objective: In this project, students will deploy and secure a virtual infrastructure composed of various interconnected components, including a Windows machine, an Ubuntu server, a FreeNAS machine, and a containerized application. The goal is to implement firewall rules, internal security services, virtual private network (VPN) configurations, and virtual local area networks (VLANs) to create a secure, segmented network architecture.

Project Description:

Students are tasked with building and securing a virtualized infrastructure that simulates a small organization's IT environment. The infrastructure will be deployed on a virtualized platform (e.g., VMware Workstation, VirtualBox, or Proxmox) and will include the following key components:

1. **Windows Machine:**

   - Deploy and configure a virtual machine (VM) running Windows (e.g., Windows 10 or Windows Server).
   - Install and configure Sysmon service.
   - Ensure that the machine is secured by configuring host-based security policies, such as Windows Defender and local firewall rules.

1. **Ubuntu Server:**

   - Deploy and configure a virtual machine running Ubuntu (e.g., Ubuntu Server 22.04).
   - Secure the server by configuring iptables, configure log collection services and enforce SSH security policies (e.g., disabling password-based authentication and enabling key-based login).

1. **FreeNAS Machine:**

   - Deploy and configure a FreeNAS VM to provide network-attached storage (NAS) services.
   - Configure shared storage with appropriate user access controls and file permissions.
   - Enable and configure security features like encryption and snapshots for data protection.

1. **Containerized Application:**

   - Set up and deploy a containerized application using Docker.

- Ensure the application runs within a secure environment and is accessible only within the private network.
- Configuring monitoring through Prometheus and Grafana.

Network and Security Configuration:

1. **Firewall (FW) Configuration and Network segmentation:**

- Deploy a virtual firewall (e.g., pfSense, OPNSense) to act as the gateway for the network.
- Configure the firewall to provide network segmentation and to enforce access control policies, restrict external access to critical services, and apply traffic filtering rules for inbound and outbound traffic.
- Implement network address translation (NAT) where necessary and create secure routing policies between different network segments.

1. **Internal Security Services:**

- Implement antivirus or endpoint protection solutions for the virtual machines.
- Configure security auditing on the Windows and Ubuntu machines to monitor for suspicious activity.
- Install and configure monitoring solution through Prometheus and Grafana.

1. **Virtual Private Network (VPN):**

- Set up a VPN (e.g., OpenVPN) on the firewall to allow secure remote access to the virtual infrastructure.
- Ensure VPN users can only access authorized resources.
- Enforce policies for secure remote access, ensuring encrypted communication across the network.

1. **Virtual Local Area Networks (VLANs):**

- Design and implement VLANs to segment the network into separate zones (e.g., separating the Windows machine, Ubuntu server, FreeNAS, and containerized application into different VLANs).
- Configure the firewall to manage inter-VLAN routing and enforce segmentation policies, limiting access between different network segments based on security needs.

Deliverables:

A Final pdf Report that includes the following sections

1. **Network Diagram**: A clear and concise diagram showing the physical and logical architecture of the virtual infrastructure, including VMs, VLANs, firewall placement, VPN endpoints, and interconnections.
2. **Configuration Documentation**: A step-by-step report including the following parts:

a. The security issues related to virtualized environment.
b. The report detailing the VM-to-VM escape workshop/
c. The installation of the FreeNas storage solution
d. The installation and configuration of the containerized application
e. The configuration of each machine, firewall rules, VPN setup, VLAN configuration, and any security measures implemented.

3. **Demonstration**: A live demonstration showing the deployment, configuration, and testing of the virtual infrastructure, focusing on key security components (firewall, VPN, VLANs, etc.).
4. **Task Assessment Table**: detailing the individuals responsible for each task and configuration.